

Snap-to-Sell: An Open-Source Pipeline for Turning a Corner-Store Product Photo into a Priced, Image-Upgraded Listing

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Abstract—Independent corner stores struggle to digitise inventory for online sale because creating a catalogue entry per product is slow and manual. We present Snap-to-Sell, an open-source pipeline that turns a single phone photo of a packaged product into a publishable listing: it recognises the item with a vision-language model fused with packaging OCR, retrieves a canonical name, price, and product image from open product catalogues and a shopping-search backend, generates grounded listing copy, replaces the shelf photo with an LLM-verified same-product catalogue image, and applies responsible-AI guardrails (age-restricted-item flagging, hallucination and low-confidence routing) before human review. We evaluate the pipeline on a hand-labelled set of 58 corner-store products spanning food, beverage, household, personal-care, tobacco, and alcohol categories, measuring identification accuracy, price error, age-restriction recall, and the image-upgrade rate against a naive vision-language caption baseline. Snap-to-Sell improves top-1 identification (86.2% vs. 84.5%) and category accuracy (79.3% vs. 77.6%), flags 100% of age-restricted items, and upgrades 29% of images, while its retrieval-grounded prices are less accurate than the baseline’s direct estimate on low-value fast-moving goods (43.1% vs. 22.4% median absolute error)—a failure mode we trace to multipack and look-alike matches and mitigate by declining to price and routing low-confidence matches to human review.

Index Terms—retail product recognition, vision-language models, e-commerce, listing generation, price estimation, responsible AI

I. INTRODUCTION AND BACKGROUND

II. PROBLEM STATEMENT AND MOTIVATION

III. RELATED WORK

Automatically turning a product photograph into a priced listing draws on four established threads: listing-text generation, fine-grained product recognition, price estimation, and product image retrieval. Recent listing-generation work favours frozen foundation models: ModICT generates descriptions from an image and keywords using a frozen language model with in-context references, and names the failure we guard against—generic, inaccurate copy for same-category products [1], [2]. Fine-grained recognition of packaged goods is well supported by large datasets such as RP2K and the smartphone-captured CAPG-GP collection [3], [4], and fusing the product image with packaging OCR outperforms image-only recognition, especially with few samples [5]. Price estimation has moved from image-only regression [6], [7]

TABLE I

PERFORMANCE ON THE 58-ITEM CORNER-STORE TEST SET. LOWER IS BETTER FOR PRICE ERROR; THE BETTER SYSTEM PER ROW IS SHOWN IN BOLD.

Metric	Baseline	Snap-to-Sell
Top-1 identification accuracy	84.5%	86.2%
Category accuracy	77.6%	79.3%
Age-restriction recall	0%	100%
Image-upgrade (swap) rate	0%	29.3%
Price median abs. % error	22.4%	43.1%
Price mean abs. % error	47.4%	359.9%

toward retrieval-grounded, confidence-aware generation [8]. Finally, the image-upgrade step rests on product visual search and same-product verification [9], [10], which also supply restricted-item detection reusable for our age-restriction guardrail.

IV. METHODOLOGY

Snap-to-Sell is a staged pipeline with a human-in-the-loop checkpoint before publication, shown in Fig. 1. The data artifacts exchanged between stages are shown in Fig. 2.

V. IMPLEMENTATION AND DESIGN

VI. RESULTS

We score the full pipeline against a naive baseline—the same recognition stage followed by ungrounded listing generation (no retrieval, image upgrade, or guardrails)—on 58 hand-labelled products, each with a real observed price. Both systems are evaluated with identical metrics; price error is the absolute percentage difference between the predicted and the observed price. Table I reports the headline figures.

Snap-to-Sell outperforms the baseline on four of the five measured axes: it recognises the exact product slightly more often, assigns the correct category more often, flags every age-restricted item (tobacco and alcohol) where the baseline flags none, and upgrades roughly one image in three to a verified same-product catalogue photo. The remaining axis, price, is the one clear failure mode: the baseline’s direct price estimate is more accurate than our retrieval-grounded price on these low-value fast-moving goods. Inspection of the per-item logs attributes this to the shopping backend returning

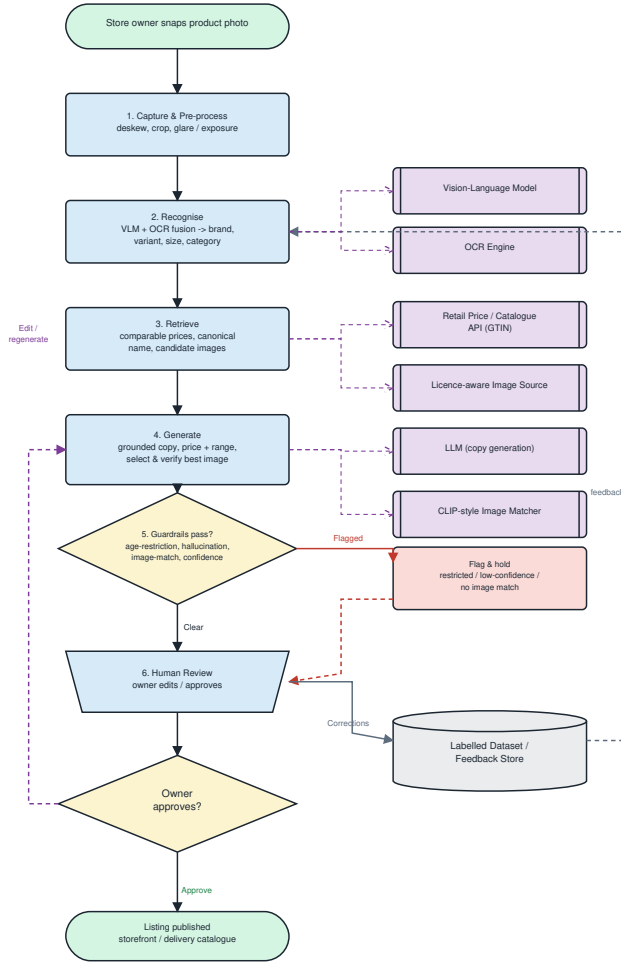


Fig. 1. End-to-end architecture: capture, recognise (VLM + OCR), retrieve (price, canonical name, candidate images), generate, guardrails, human review, and publish, with a feedback loop to a labelled store.

multipack or case listings and visually similar look-alikes for cheap single-unit items, so a plausible-looking but wrong comparable anchors the price. The confidence gate mitigates the worst cases by declining to price and routing to review the eight items whose retrieved brand could not be confirmed; the reported price figures are computed over the items each system did price, so the two price columns cover slightly different subsets. Top-3 accuracy and a blind human listing-quality rating were not automated for this run and are left to future work.

VII. CONCLUSION AND FUTURE WORK

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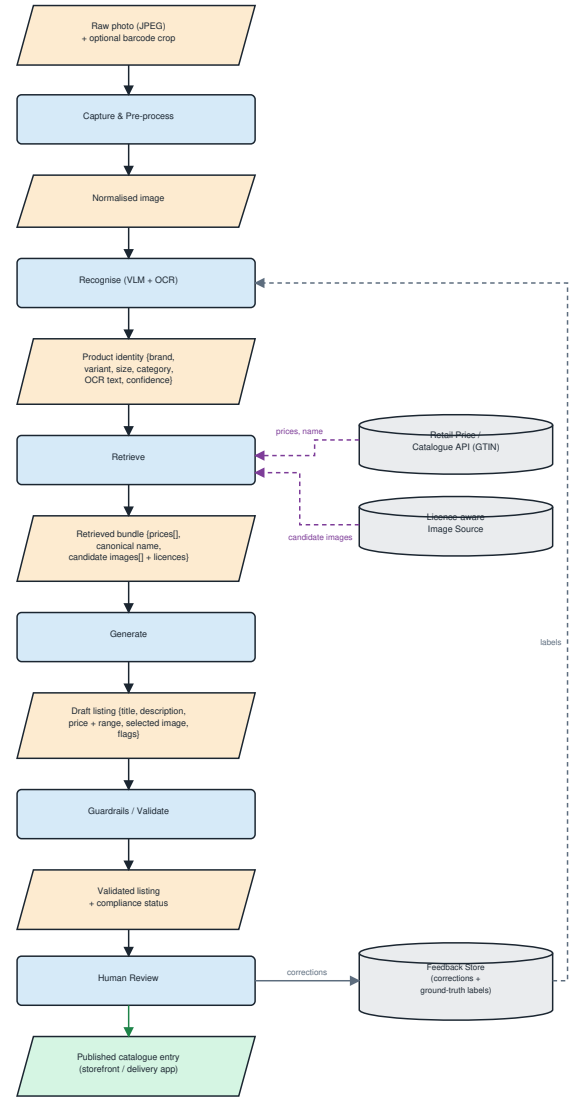


Fig. 2. Data-flow view: the artifacts passed between stages, from raw photo to published catalogue entry, with external datastores feeding retrieval.

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